



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

SCENARIO:

You are a cybersecurity analyst working for a multimedia company that offers web design services, graphic design, and social media marketing solutions to small businesses. Your organization recently experienced a DoS attack, which compromised the internal network for two hours until it was resolved.

During the attack, your organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Normal internal network traffic could not access any network resources. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services.

The company's cybersecurity team then investigated the security event. They found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a denial of service (DoS) attack.

To address this security event, the network security team implemented:

- A new firewall rule to limit the rate of incoming ICMP packets
- Source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets
- Network monitoring software to detect abnormal traffic patterns
- An IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics

As a cybersecurity analyst, you are tasked with using this security event to create a plan to improve your company's network security, following the National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF). You will use the CSF to help you navigate through the different steps of analyzing this cybersecurity event and integrate your analysis into a general security strategy. We have broken the analysis into different parts in the template below. You can explore them here:

- **Identify** security risks through regular audits of internal networks, systems, devices, and access privileges to identify potential gaps in security.
- **Protect** internal assets through the implementation of policies, procedures, training and tools that help mitigate cybersecurity threats.
- **Detect** potential security incidents and improve monitoring capabilities to increase the speed and efficiency of detections.
- **Respond** to contain, neutralize, and analyze security incidents; implement improvements to the security process.

Recover affected systems to normal operation and restore systems data and/or assets that have been affected by an incident.

Summary	A multimedia company recently experienced an attack that rendered the internal network compromised for two hours. It was resolved by the incident response management team by blocking incoming ICMP packets to restore critical network infrastructure. The cybersecurity team discovered an unconfigured firewall that allowed the malicious hacker to overload the network services through a Denial-of-Service (DoS) attack. The network security team implemented the proper configurations in the firewall, software, and other systems.
Identify	A DOS attack happened at a multimedia company where network was down for two hours due to ICMP packet flood done by a malicious actor. This was identified by the cybersecurity team.
Protect	To prevent offline network services, the incident response management team blocked incoming ICMP packets to stop offline and restore critical network services. The network security team added rule/s in the firewall such as limiting the rate of incoming ICMP packets and IDS/IPS system filter based on suspicious characteristics and behavior.
Detect	The network security team implemented source IP address verification for spoofed IP addresses in the firewall as well as implement network monitoring software for detecting abnormal traffic patterns.
Respond	The cybersecurity team will isolate affected systems so as to prevent downtime in a network. Further investigations and analysis of network logs is required to check abnormal activities and report to upper management or authorities if needed.
Recover	Since the network is up and running with the added configurations, the team documented the attack as reference for future security breaches and mitigations for prevention.

Reflections/Notes: The unconfigured firewall rule is always present in a company network. To prevent this from happening, weekly checkups of firewall configurations and other network infrastructure systems is a must to help strengthen security.